

Section A

- 1 (a) force proportional to product of masses and inversely proportional to
square of separation (*do not allow square of distance/radius*) M1
either point masses *or* separation @ size of masses A1 [2]
- (b) (i) $\omega = 2\pi / (27.3 \times 24 \times 3600)$ or $2\pi / (2.36 \times 10^6)$ M1
 $= 2.66 \times 10^{-6} \text{ rad s}^{-1}$ A0 [1]
- (ii) $GM = r^3 \omega^2$ or $GM = v^2 r$ C1
 $M = (3.84 \times 10^5 \times 10^3)^3 \times (2.66 \times 10^{-6})^2 / (6.67 \times 10^{-11})$ M1
 $= 6.0 \times 10^{24} \text{ kg}$ A0 [2]
(special case: uses $g = GM/r^2$ with $g = 9.81$, $r = 6.4 \times 10^6$ scores max 1 mark)
- (c) (i) grav. force $= (6.0 \times 10^{24}) \times (7.4 \times 10^{22}) \times (6.67 \times 10^{-11}) / (3.84 \times 10^8)^2$ C1
 $= 2.0 \times 10^{20} \text{ N}$ (*allow 1 SF*) A1 [2]
- (ii) *either* $\Delta E_p = Fx$ because F constant as $x \ll$ radius of orbit B1
 $\Delta E_p = 2.0 \times 10^{20} \times 4.0 \times 10^{-2}$ C1
 $= 8.0 \times 10^{18} \text{ J}$ (*allow 1 SF*) A1 [3]
- or* $\Delta E_p = GMm/r_1 - GMm/r_2$ C1
Correct substitution B1
 $8.0 \times 10^{18} \text{ J}$ A1
($\Delta E_p = GMm/r_1 + GMm/r_2$ is incorrect physics so 0/3)

- 2 (a) energy $= \frac{1}{2} m \omega^2 a^2$ and $\omega = 2\pi f$ C1